# 24-HOUR DIGITAL CLOCK
# CHRONOBITS'26

**This project implements a 24-Hour Digital Clock using the NE555 Timer and CD4026 Counter ICs without a microcontroller. The NE555 generates a 1 Hz clock pulse, while cascaded CD4026 ICs count the pulses and drive six seven-segment displays to show time in HH:MM:SS format. External reset logic is used to achieve MOD-60 and MOD-24 counting for accurate timekeeping.**

**Objectives**

- **Design a 24-hour digital clock using discrete ICs.**
- **Generate a 1 Hz clock pulse using the NE555 Timer.**
- **Display time in HH:MM:SS format using CD4026 ICs.**
- **Implement MOD-60 and MOD-24 counting using reset logic.**


**Team Name : THE ASTABLE MINDS**

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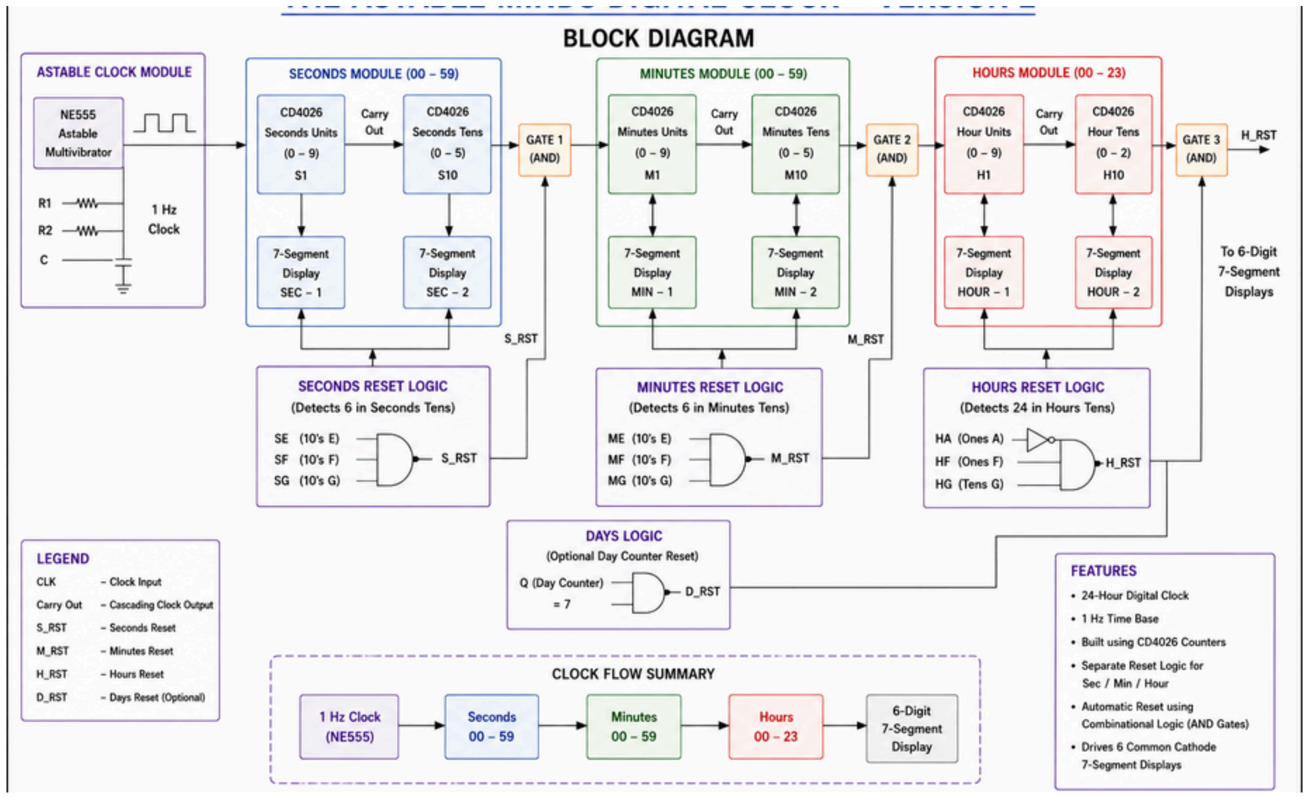
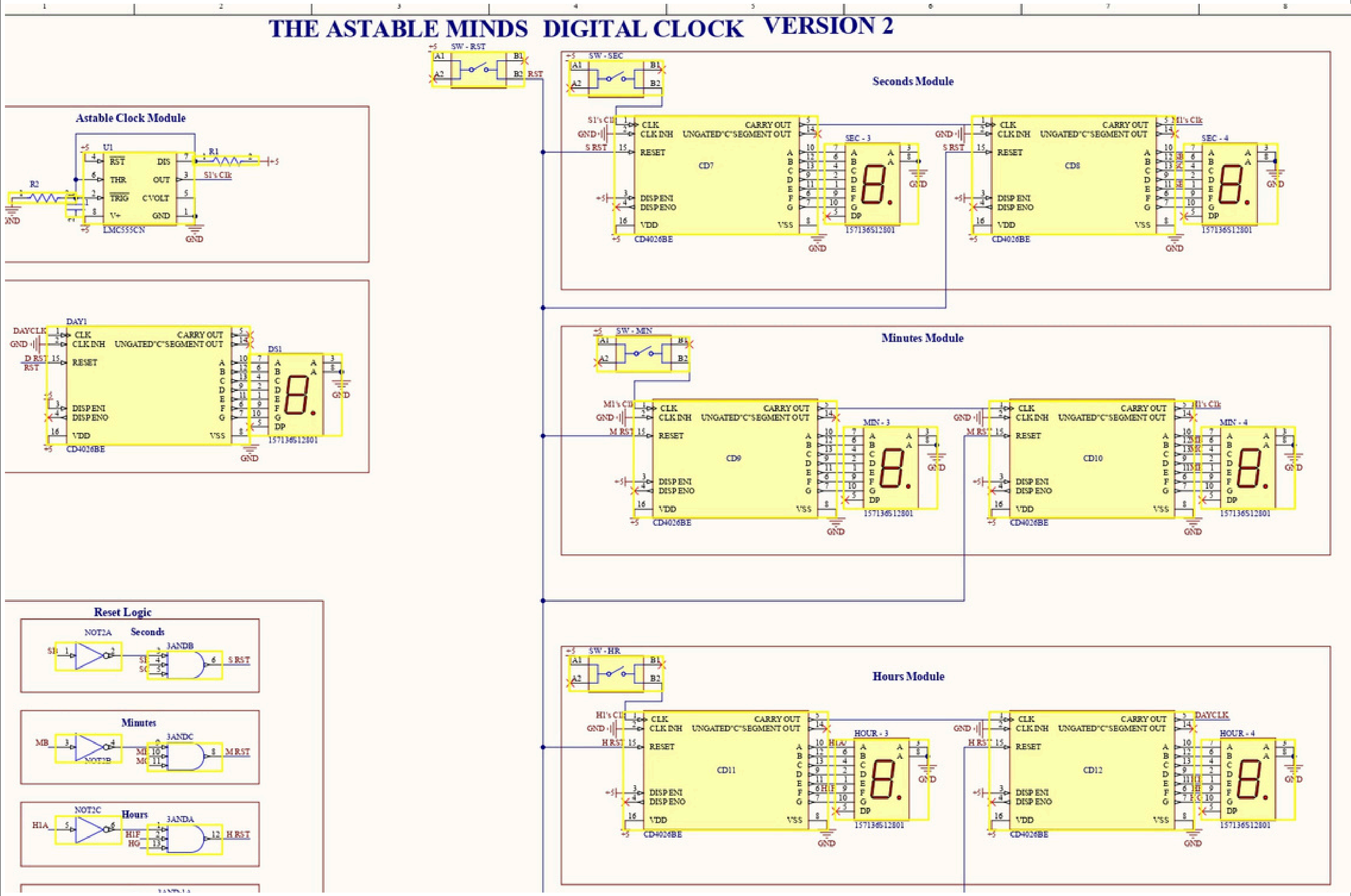


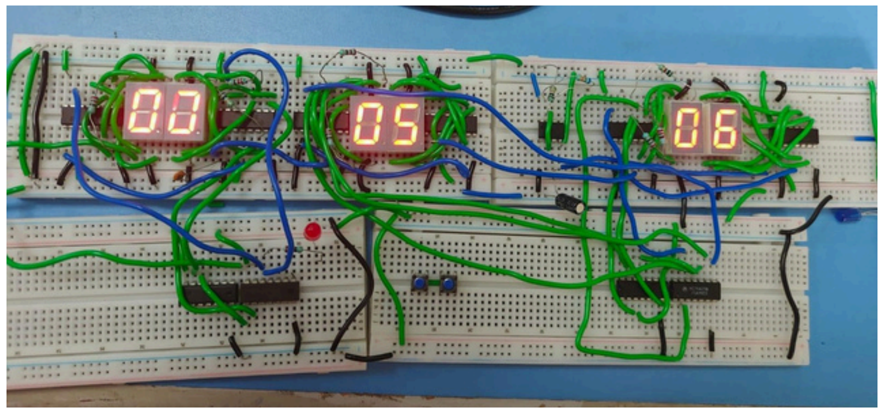




## Working Principle

1. **The NE555 Timer is configured in astable mode to generate a continuous 1 Hz clock pulse.**
2. **Each pulse is applied to the first CD4026 counter, which increments the seconds display.**
3. **The Carry Out output cascades the count to the next CD4026, forming the seconds, minutes, and hours counters.**
4. **External reset logic detects the count of 60 for seconds and minutes, resetting them to 00 (MOD-60).**
5. **Additional logic detects 24 hours and resets the hour counters to 00, achieving 24-hour operation.**
6. **The time is displayed in HH:MM:SS format using six common-cathode seven-segment displays.**
7. **This version is concise, technically accurate, and fits well on an A3 engineering poster.**

## Results

- **Successfully displays time in HH:MM:SS**
- **Correct MOD-60 counting**
- **Correct MOD-24 rollover**
- **Stable 1 Hz pulse generation**



| Component | Qty |
| --- | --- |
| NE555 | 1 |
| CD4026 | 6 |
| 7-Segment Display | 6 |
| 74HC04 | 1 |
| 74HC11 | 1 |